

TUFT as an Infinite Registry and Topological Quantum Error Correction in Fibonacci Microtubule Pathways

A formal response to Stuart Hameroff's question

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1 Executive response

Yes. This line of inquiry deserves direct development, and it intersects existing Orch OR work more closely than a generic proposal to apply quantum error correction to biology.

The immediate point of contact is the proposal that microtubules contain intersecting left- and right-handed helical pathways indexed by the Fibonacci sequence 3, 5, 8, 13, ..., together with the 2025 *BioSystems* paper by An and Choi that maps those pathways to a surface-code architecture. Their paper treats asymmetric Fibonacci helical pathways as candidate logical-qubit structures and asks whether their intersections could stabilize a surface code. A 2026 follow-up analyzes the feasibility conditions under which such a code might extend coherence to times relevant to Orch OR.

The contribution offered here is different and complementary. It begins from Jenny Lorraine Nielsen's **Topological Unified Field Theory (TUFT)**, not because acceptance of TUFT as a theory of everything is required, but because the registry studies establish a weaker and independently useful result: **TUFT functions as an adequate infinite registry**. That result matters directly to error correction. An infinite registry supplies a carrier broad enough to represent admissible local states, global paths, conjugate sectors, topological defects, coverings, quotients, operator

spectra, and reattachment maps. Quantum error correction can then be formulated as the disciplined selection of a protected subregistry and the management of the residue that remains after local repair.

The proposed research claim should therefore remain precise:

The TUFT registry can host an explicit topological-code model of the Orch OR microtubule without requiring prior acceptance of TUFT’s full ontological claims. The decisive question is whether the independently specified Fibonacci-helical microtubule geometry selects a sparse, physically admissible code subspace in which local disturbances produce syndromes while a nonlocal logical class survives local re-description and repair.

This formulation places quantum error correction where it belongs in the Lambda / NSAF / TUFT canon: as a **residue problem**.

2 1. The question in its proper Orch OR context

Stuart Hameroff’s question was:

“Have you looked at topological quantum error correction along microtubule lattice pathways following Fibonacci geometry? Maybe we should.”

Because this question comes from the principal biological architect of Orch OR, an adequate answer should not rehearse the elementary claim that microtubules may process information. Hameroff’s work already supplies that premise. In the 2002 study on conduction pathways, Hameroff, Nip, Porter, and Tuszynski treated microtubules as cylindrical tubulin lattices capable of signaling, conductivity, and classical or quantum information processing. The 2014 Orch OR review then placed coherent microtubule processes within a theory in which organized quantum evolution terminates through objective reduction.

The relevant overlap therefore begins one level later:

1. Orch OR supplies a proposed physical carrier: the microtubule and its tubulin pathways.
2. The Fibonacci-helical work supplies a candidate lattice organization: intersecting 3-, 5-, 8-, and 13-start pathways.
3. Surface-code work supplies a candidate protection mechanism: local syndrome detection with nonlocal logical storage.
4. TUFT supplies an adequate infinite registry in which these path systems, coverings, conjugate sectors, defects, operators, and residual classes can be represented together.
5. The Lambda residue formalism supplies the criterion separating correctable local variation from irreducible logical change.

The answer is therefore not merely “yes, this resembles a surface code.” It is:

Yes. The Orch OR Fibonacci pathway proposal already has a first surface-code realization. TUFT and the residue formalism can sharpen that realization by defining what counts as the code space, what counts as a local error, what the syndrome mea-

tures, which transformations remain admissible, and which nonlocal class must survive reattachment.

3 2. TUFT's pertinent status: adequate infinite registry

3.1 2.1 The weaker result is sufficient

The error-correction argument does not require the following strong premise:

TUFT is the final correct theory of all physical reality.

It requires only:

TUFT is an adequate registry for the relevant topological and spectral relations.

Call the complete registry $\mathfrak{R}_\infty$. Its adequacy means that it can represent, at minimum:

- circle-fibered and toroidal carriers;
- discrete quotients and covering spaces;
- left/right or conjugate phase sectors;
- holonomy and winding data;
- local operators and their spectra;
- defects, seams, and gluing obstructions;
- projection, descent, and norm-like recomposition;
- finite code subspaces selected by additional constraints.

This is already enough to formulate a code. A code does not need the registry to determine all of physics; it needs the registry to distinguish physical states, error sectors, logical equivalence classes, and recovery maps without contradiction.

3.2 2.2 Infinite registry does not itself provide a derivation

The registry result must be used with the canon's own warning. If $\mathfrak{R}_\infty$ contains every admissible path, then demonstrating that a Fibonacci surface-code path occurs somewhere inside it carries little information. The meaningful object is a constrained reachable sector.

Let $\mathcal{G} = \{g_1, \dots, g_m\}$ be independently fixed generators and x_0 a physical initial state. Define

$$\text{Reach}_{\mathcal{G}}(x_0; N) = \{g_{i_k} \cdots g_{i_1} x_0 : k \leq N\}.$$

A candidate mechanism gains evidentiary force only when:

- the generators arise from measured microtubule geometry or dynamics;
- the allowed path length or operation count remains bounded;
- the code construction is sparse rather than arbitrarily selectable;

- control geometries fail under the same rules;
- the model predicts a measurable threshold, lifetime, spectrum, or defect response.

Thus TUFT serves two logically different functions:

$$\boxed{\mathfrak{R}_\infty \text{ supplies representational completeness}}$$

and

$$\boxed{\mathcal{G} \text{ supplies explanatory constraint.}}$$

Quantum error correction concerns the second function operating inside the first.

4 3. Quantum error correction as a residue problem

4.1 3.1 Standard code condition

Let $\mathcal{C} \subset \mathcal{H}$ be a code space with projector P , and let $\{E_a\}$ be a family of correctable errors. The Knill-Laflamme condition states

$$PE_a^\dagger E_b P = c_{ab} P.$$

The condition means that the environment or syndrome apparatus may distinguish error sectors without distinguishing the encoded logical state. Local disturbance becomes visible, while logical information remains hidden in an equivalence class.

The code therefore separates:

$$\text{physical variation} = \text{correctable component} + \text{logical residue}.$$

This is structurally equivalent to the canon's obstruction decomposition

$$\mathcal{O} = \mathcal{A} + \mathcal{J},$$

where $\mathcal{A}$ denotes the representation-dependent or removable component and $\mathcal{J}$ denotes the orbit-minimized irreducible residue.

4.2 3.2 Chain-complex formulation

In a homological surface code, an error occupies a chain e . Its syndrome is the boundary

$$s = \partial e.$$

A decoder chooses a correction chain c satisfying

$$\partial c = s.$$

The combined chain is closed:

$$\partial(e + c) = 0.$$

Yet correction succeeds only if that closed chain is homologically trivial:

$$[e + c] = 0.$$

A logical error occurs when

$$[e + c] \neq 0.$$

This makes the residue interpretation exact rather than metaphorical. The syndrome identifies the local boundary of the disturbance. The decoder removes a locally reparable representative. The nonzero homology class that may remain constitutes the logical residue.

Equivalently:

$$\boxed{\text{syndrome} = \text{observable boundary}}$$

$$\boxed{\text{recovery} = \text{admissible re-description}}$$

$$\boxed{\text{logical failure} = \text{nontrivial residue after reattachment.}}$$

4.3 3.3 Residue functional for a biological code

Let x denote the state of a microtubule registry, $R(x)$ the set of physically admissible local repair operations, and T a target logical sector. Define

$$\mathcal{J}(x; T) = \inf_{r \in R(x)} \mathcal{O}(r, x; T).$$

Possible obstruction functionals include

$$\mathcal{O}_{\text{logical}} = 1 - F(\rho_L, (\mathcal{R} \circ \mathcal{E})(\rho_L)),$$

$$\mathcal{O}_{\text{leakage}} = \text{Tr}(P_{\perp}\rho),$$

or, for a topological code,

$$\mathcal{O}_{\text{top}} = \mathbf{1}_{[e+c] \neq 0}.$$

The admissible set $R(x)$ must remain biologically constrained. It may include local conformational relaxation, tubulin replacement, phosphorylation-state change, ionic redistribution, motor- or MAP-mediated correction, or collective phase relaxation. It may not include arbitrary global rewriting. Otherwise the residue vanishes by definition and the model loses physical content.

5 4. Lift-twist-reattach as the code operation

TUFT's lift-twist-reattach operation provides a natural registry grammar for error correction.

5.1 4.1 Lift: resolve hidden sectors

A noisy state is lifted into a larger syndrome-resolved carrier:

$$\mathcal{H}_{\text{physical}} \longrightarrow \mathcal{H}_{\text{logical}} \otimes \mathcal{H}_{\text{syndrome}}.$$

At the microtubule level, the lift may correspond to resolving a local tubulin or pathway state into:

- its logical-path membership;
- local parity or phase information;
- seam-relative orientation;
- left- versus right-handed helical sector;
- neighboring-check outcomes.

The lift restores distinctions hidden by the coarse biological description.

5.2 4.2 Twist: error, holonomy, or conjugate transport

An error changes local phase, occupation, parity, conformation, or transport sector:

$$|\psi_L\rangle|0_s\rangle \mapsto E_a|\psi_L\rangle|s_a\rangle.$$

In the Orch OR geometry, the most immediate candidate twist is the relation between intersecting left- and right-handed Fibonacci helices. A physically adequate model must define this twist as an operator, not merely as visual chirality. Candidates include:

- a phase-conjugation map;
- a seam-crossing holonomy;
- a parity exchange between helical families;

- a local braid or path-exchange operator;
- an error-induced transition between pathway sectors.

5.3 4.3 Reattach: decode without erasing the logical class

Recovery uses syndrome information to return the state to the code space:

$$\mathcal{R}_a E_a |\psi_L\rangle \approx |\psi_L\rangle.$$

The reattachment succeeds when locally altered pieces recombine without changing the global winding or logical class. It fails when the chosen correction closes into a noncontractible loop or otherwise changes the protected sector.

This yields a third exact realization of lift-twist-reattach:

$\text{syndrome lift} \rightarrow \text{error-sector twist} \rightarrow \text{code-space reattachment}$

6 5. Why the Fibonacci microtubule geometry is not incidental

6.1 5.1 Existing Orch OR pathway proposal

The relevant proposal treats tubulin-level information as organized through intersecting helical pathways in both handed directions, with pathway counts following the series

$$3, 5, 8, 13, \dots$$

rather than through a merely additive list of independent tubulin bits. An and Choi’s 2025 model explicitly interprets asymmetric Fibonacci helical pathways as logical qubits within a surface-code architecture. Their work therefore already identifies the core structural overlap Hameroff’s question asks about.

6.2 5.2 Fibonacci geometry can play several distinct roles

The term “Fibonacci” should not remain semantically ambiguous. It may denote:

1. **Helical winding counts:** geometrically distinct paths around the microtubule cylinder.
2. **A substitution sequence:** $A \mapsto AB$, $B \mapsto A$, producing quasiperiodic couplings.
3. **A spectral hierarchy:** gaps, mode counts, or transfer amplitudes organized by Fibonacci recursion.
4. **A Fibonacci anyon category:** non-Abelian fusion obeying $\tau \times \tau = 1 + \tau$.
5. **A registry grammar:** recursive closure under two prior states.

These roles should not be conflated. A microtubule may possess Fibonacci helical path counts without hosting Fibonacci anyons. Conversely, a Fibonacci-modulated coupling network may produce useful spectral or localization properties without implementing a topological quantum code.

6.3 5.3 The strongest defensible initial claim

The strongest claim presently justified is:

The intersecting Fibonacci-helical microtubule geometry supplies a nontrivial candidate graph for a surface or subsystem code, and TUFT supplies an adequate infinite registry in which its left/right sectors, defects, coverings, holonomies, and logical residues can be represented. Whether biological microtubules physically instantiate the required quantum degrees of freedom, measurements, and recovery dynamics remains an experimental question.

7 6. A concrete TUFT-Orch OR code model

7.1 6.1 Physical graph

Represent a microtubule as a cylindrical graph

$$G = (V, E_L \cup E_R \cup E_C),$$

where:

- V represents tubulin dimers or coarse-grained tubulin clusters;
- E_L represents left-handed Fibonacci-helical couplings;
- E_R represents right-handed Fibonacci-helical couplings;
- E_C represents local circumferential or protofilament couplings;
- a distinguished seam Σ records the closure mismatch or branch cut.

Assign a local Hilbert space $\mathcal{H}_v \cong \mathbb{C}^d$ to each $v \in V$. The full carrier is

$$\mathcal{H} = \bigotimes_{v \in V} \mathcal{H}_v.$$

A minimal model may start with qubits, $d = 2$, while a more biologically faithful model may require qudits representing multiple conformational or dipolar states.

7.2 6.2 Checks and logical operators

Define local check operators on elementary cells or intersections:

$$A_s = \prod_{v \in \text{star}(s)} X_v, \quad B_p = \prod_{v \in \partial p} Z_v.$$

The code space satisfies

$$A_s |\psi\rangle = |\psi\rangle, \quad B_p |\psi\rangle = |\psi\rangle.$$

Because the physical graph is cylindrical and contains intersecting helical families, candidate logical operators include:

$$\bar{Z} = \prod_{v \in \gamma_{\text{circ}}} Z_v,$$

around a noncontractible circumferential path, and

$$\bar{X} = \prod_{v \in \gamma_{\text{helix}}} X_v,$$

along a seam-relative or longitudinal helical path. More sophisticated constructions could encode logical information in relative winding between E_L and E_R rather than in either family alone.

7.3 6.3 Seam as a mathematically active object

The seam should not be treated only as an imperfection. It may serve one of four roles:

- a branch cut required to define global phase consistently;
- a boundary supporting modified stabilizers;
- a syndrome line at which path parity changes;
- a biologically weak locus that increases rather than suppresses error.

The fourth possibility must remain part of the model. The seam becomes useful only if its induced check structure improves logical protection or supplies a measurable syndrome channel.

7.4 6.4 Noise model

A biologically relevant noise channel should include more than independent Pauli errors. A first hierarchy might include:

$$\mathcal{E} = \mathcal{E}_{\text{deph}} \circ \mathcal{E}_{\text{relax}} \circ \mathcal{E}_{\text{loss}} \circ \mathcal{E}_{\text{corr}},$$

where:

- $\mathcal{E}_{\text{deph}}$ models phase noise from the thermal and ionic environment;
- $\mathcal{E}_{\text{relax}}$ models conformational or dipolar relaxation;
- $\mathcal{E}_{\text{loss}}$ models missing, replaced, or chemically altered tubulin units;
- $\mathcal{E}_{\text{corr}}$ models spatially correlated disturbances along protofilaments or helical pathways.

A surface-code interpretation gains force only if it tolerates correlated noise appropriate to the microtubule rather than only an idealized independent-error channel.

7.5 6.5 Recovery mechanism

A biological code cannot assume an external cryogenic controller unless the proposal is explicitly a quantum simulation of the microtubule rather than a mechanism occurring in vivo. For an in-vivo

Orch OR model, recovery must arise from endogenous processes, possibly:

- driven-dissipative relaxation into a protected manifold;
- tubulin turnover and lattice self-repair;
- microtubule-associated proteins acting as local checks;
- CaMKII or other phosphorylation patterns acting as write/read constraints;
- motor-protein traffic supplying repeated parity-sensitive interactions;
- resonant collective modes suppressing off-code excitations.

These remain hypotheses, but they define experimentally addressable mechanisms.

8 7. TUFT-specific value added

8.1 7.1 Registry equivalence without ontological collapse

TUFT permits several readings of a common carrier through different operators. For this project, that means the same microtubule registry can be studied through:

- graph Laplacians for transport and diffusion;
- Dirac-type operators for chiral or spinorial couplings;
- Beltrami operators for helical flow and handedness;
- stabilizer Hamiltonians for code protection;
- transfer matrices for Fibonacci pathway propagation;
- cohomology for global obstruction classes.

No single operator must be declared foundational. The code is a structured reading of the carrier.

8.2 7.2 Residue rather than perfect isolation

Orch OR has often been challenged on the basis of environmental decoherence. Error correction changes the form of the question. The relevant requirement is not perfect preservation of every microscopic phase. It is preservation of the encoded logical class under the expected local noise channel.

Thus the decisive quantity is not simply the bare coherence time T_2 , but the logical lifetime

$$T_L = T_L(d, p, \xi, \tau_{\text{cycle}}, \mathcal{R}),$$

as a function of code distance d , physical error rate p , correlation length ξ , correction or relaxation cycle τ_{cycle} , and recovery process $\mathcal{R}$.

The 2026 feasibility study moves in this direction by comparing surface-code-supported coherence against times required by Orch OR. TUFT's residue formalism adds a precise interpretation: logical lifetime measures how long the global class remains invariant after minimization over admissible local repairs.

8.3 7.3 Operator-agnostic comparison of candidate encodings

TUFT can compare several code architectures on the same physical registry:

- ordinary surface code;
- subsystem or gauge code adapted to uncertain checks;
- Floquet code adapted to periodically changing interactions;
- bosonic or oscillator code on collective vibrational modes;
- non-Abelian code if fusion-like excitations can be physically identified;
- classical topological memory as a control model.

This avoids prematurely identifying the Fibonacci pathway with Fibonacci anyons. The registry can host both possibilities while experiments determine which operator reading, if any, the microtubule realizes.

9 8. What must be demonstrated for a valid claim

A rigorous claim that Orch OR microtubules implement topological quantum error correction requires all of the following.

9.1 8.1 Physical information-bearing degree of freedom

Specify the qubit or qudit:

- tubulin conformation;
- electric dipole orientation;
- tryptophan excitation;
- collective vibrational mode;
- ionic or ordered-water state;
- another explicitly measured two- or multi-level subsystem.

9.2 8.2 Code subspace

Define

$$\mathcal{C} \subset \mathcal{H}$$

and identify the physical constraints that stabilize it.

9.3 8.3 Check operators or autonomous equivalent

Identify local observables whose outcomes reveal errors without revealing logical information. If no explicit measurements occur, derive an autonomous dissipative mechanism equivalent to repeated syndrome extraction.

9.4 8.4 Logical operators

Define nonlocal operators $\overline{X}$ and $\overline{Z}$, or their non-Abelian analogues, in terms of measurable microtubule pathways.

9.5 8.5 Noise and recovery

Provide a biologically realistic channel $\mathcal{E}$ and a physical recovery process $\mathcal{R}$.

9.6 8.6 Scaling or threshold

Show that increasing the effective code distance improves logical protection, for example

$$p_L \sim e^{-\alpha d}$$

below a threshold, or derive another nontrivial suppression law appropriate to correlated biological noise.

9.7 8.7 Exclusionary controls

Compare the Fibonacci-helical geometry against:

- periodic helical pathways;
- random pathway assignments;
- Thue-Morse or other quasiperiodic sequences;
- seam-free and seam-modified cylinders;
- optimized non-Fibonacci graphs with the same number of edges and parameters.

The Fibonacci model becomes informative only if the same fixed parameter budget fails to reproduce its protection in controls.

10 9. A staged research program

10.1 Stage I: exact combinatorial reconstruction

Construct the microtubule pathway graph from the proposed 3-, 5-, 8-, and 13-start helices. Determine:

- intersections;
- cycle basis;
- homology;
- seam dependence;
- graph distance;
- shortest noncontractible logical paths;
- whether left/right pathway families generate independent logical sectors.

10.2 Stage II: code search inside the TUFT registry

Search for low-weight local checks compatible with the graph. The registry may contain many possible codes; impose the structured-map criterion by minimizing:

$$\mathcal{L} = \lambda_1(\text{check weight}) + \lambda_2(\text{nonlocal control cost}) + \lambda_3(\text{biophysical implausibility}) - \lambda_4(\text{code distance}).$$

10.3 Stage III: realistic noise simulation

Simulate thermal dephasing, correlated pathway noise, missing tubulin units, seam-localized defects, and time-dependent couplings. Compare decoders and autonomous recovery models.

10.4 Stage IV: physical observables

Derive signatures that distinguish an error-correcting phase from generic resonance or structural robustness. Candidate observables include:

- parity-dependent response at pathway intersections;
- nonlocal recovery after localized perturbation;
- threshold-like change in lifetime versus defect density;
- seam-dependent logical switching;
- scaling of coherence or transport with pathway length;
- error syndromes correlated across left/right Fibonacci helices.

10.5 Stage V: Orch OR coupling

Only after an explicit protected logical sector is established should the model be connected to objective reduction. Then ask whether the protected state's mass-distribution difference, lifetime, and spatial extent satisfy the relevant OR condition

$$\tau \sim \frac{\hbar}{E_G}.$$

Error correction would preserve the orchestrated state up to the proposed reduction threshold; it would not replace the OR mechanism.

11 10. Tier discipline

11.1 Tier I - established or directly checkable

- Orch OR proposes coherent quantum processes in neuronal microtubules linked to objective reduction.
- Microtubules form cylindrical tubulin lattices with helical pathways and structural defects.
- Surface codes protect logical information through local syndromes and nonlocal equivalence classes.

- Homological code failure can be written as a nontrivial residual class $[e + c] \neq 0$.
- An and Choi published a 2025 surface-code model for Fibonacci helical pathways of the Orch OR microtubule.
- A 2026 follow-up analyzed feasibility conditions for extending coherence through that model.
- TUFT's registry studies provide a formal infinite registry adequate to represent the relevant carrier, operator, quotient, path, and residue structures.

11.2 Tier II - structural correspondence

- Topological QEC instantiates lift-twist-reattach as syndrome lift, error-sector twist, and code-space recovery.
- The logical class functions as an irreducible residue under admissible local corrections.
- Intersecting left/right Fibonacci helices provide a candidate dual-lattice structure for local checks and nonlocal logical paths.
- The microtubule seam may function as a branch cut, syndrome boundary, or modified-check locus.
- TUFT's operator-agnostic registry permits comparison of multiple code types without identifying them prematurely.

11.3 Tier III - open physical program

- Physiological microtubules maintain a quantum code space for behaviorally relevant times.
- Tubulin, dipolar, tryptophan, vibrational, or water-coupled states supply usable qubits or qudits.
- Biological processes perform syndrome extraction or autonomous recovery.
- Fibonacci pathway geometry produces a threshold superior to control geometries.
- The protected logical state participates in Orch OR objective reduction.
- Microtubules host Fibonacci anyons or a non-Abelian topological phase.

12 11. Formal response suitable for Stuart Hameroff

Yes. The question now has a direct starting point: An and Choi's 2025 *BioSystems* paper already models the intersecting left- and right-handed 3-, 5-, 8-, and 13-start Orch OR microtubule pathways as a surface-code architecture, and their 2026 follow-up asks whether such coding could extend coherence to Orch OR-relevant times.

Our contribution would begin from Jenny Lorraine Nielsen's Topological Unified Field Theory. Acceptance of TUFT as a theory of everything is not required for this purpose. The registry studies establish the weaker result actually needed here: TUFT provides an adequate infinite registry for the relevant fibers, helical paths, conjugate sectors, coverings, defects, holonomies, operators, and descent maps.

The key caution is that an infinite registry can contain many possible codes, so mere representability proves little. The task is to show that the independently specified microtubule generators - cylindrical closure, seam, left/right helicity, and Fibonacci

3, 5, 8, 13 pathways - select a sparse code subspace and an admissible recovery process that competing geometries do not reproduce.

In our Lambda / NSAF / TUFT formulation, quantum error correction is a residue problem. A local error chain e produces a syndrome ∂e . A decoder chooses c with $\partial c = \partial e$, leaving a closed chain $e + c$. Recovery succeeds when $[e + c] = 0$ and fails when a nontrivial global class remains. The surviving homology, winding, fusion, or holonomy class is the logical residue after all admissible local repair.

This also gives a precise lift-twist-reattach realization: lift the microtubule state into logical and syndrome sectors; let local phase, parity, conformational, or pathway errors supply the twist; then reattach through a physical decoder or autonomous relaxation mechanism while preserving the nonlocal logical class.

The next useful step would be to reconstruct the exact dual-helical microtubule graph, define candidate stabilizers at pathway intersections, identify seam-relative logical operators, impose biologically realistic correlated noise, and compare Fibonacci geometry against periodic, random, and non-Fibonacci controls. If logical lifetime scales with code distance or shows a genuine threshold under those fixed rules, the model would supply Orch OR with an explicit protection mechanism rather than requiring microscopic coherence to remain perfectly isolated.

We should therefore pursue it, but as an explicit code-construction and exclusion test: not simply asking whether TUFT can represent the idea, but whether actual microtubule geometry forces a short path to protected nonlocal information.

13 12. Conclusion

Stuart Hameroff's question identifies a valuable convergence point. Orch OR already proposes microtubules as organized quantum-information carriers. Fibonacci-helical pathway models already suggest a dual lattice. Surface-code work already supplies a first error-correction mapping. TUFT adds a complete registry in which those elements can be placed without requiring acceptance of its strongest ontological claims. Lambda residue theory then identifies the protected object: not every microscopic state, but the nonlocal logical class that remains invariant under a bounded family of local corrections.

The resulting program has a clear stopping condition. It succeeds only if the Fibonacci microtubule geometry yields an explicit code space, local checks or autonomous recovery, a realistic noise model, nonlocal logical operators, and superior threshold or lifetime behavior against controls. Until then, the correspondence remains structurally strong but physically open.

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